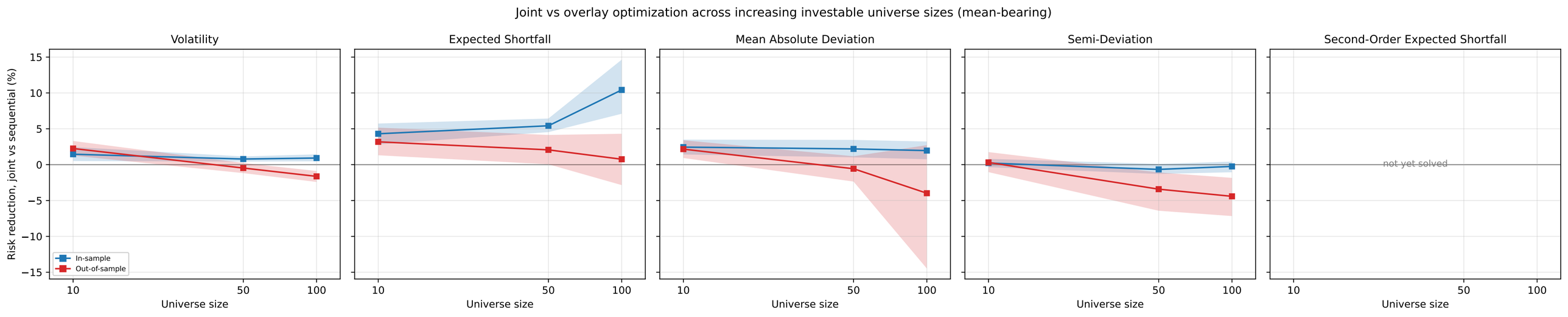
Joint vs overlay optimization across increasing investable universe sizes (mean-bearing)
Volatility
Expected Shortfall
Mean Absolute Deviation
Semi-Deviation
Second-Order Expected Shortfall
Risk reduction, joint vs sequential (%)
15
10
5
0
−5
−10
−15
In-sample
Out-of-sample
not yet solved
10
50
100
Universe size